The Idiosyncrasies of Semiconductor Contracts

A field guide to who carries the risk — read from real, SEC-filed agreements. Grounded in public exhibits; quotations verbatim; bracketed marks like [*] and [***] are the issuer's own redactions. Genuine advice, not legal advice.

Every chip company signs a version of the same three deals — with a **foundry** to make its parts, a **customer** to buy them, and a **licensor** for the IP underneath — and carries a fourth exposure nobody signs at all: **who ultimately owns the counterparty.** The terms are filed as exhibits to SEC registration statements and annual reports. They are public, and almost never read.

Generic contract review treats these like any commercial agreement. They are not. Semiconductor supply has its own grammar — the wafer bank, the yield floor, the process-learning firewall, take-or-pay against a *non-binding* forecast, the change-of-control tripwire hiding under a foreign parent. The dollar figures are usually redacted. **The structure never is** — and the structure is where the risk lives.

This guide is organized the way a buyer or a supplier actually experiences risk: **by dimension, and by who carries it.** The counts don't matter here; the *patterns* do, and across the agreements we've read they are strikingly consistent. Read it before you sign — because the party across the table has signed this many times, and you are signing it once.

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1 · Volume & commitment — who eats the forecast

The idiosyncrasy. In most industries a purchase order is a purchase order. In semiconductors the *forecast* and the *commitment* are deliberately split: a buyer gives a non-binding rolling forecast, then the contract quietly makes some window of it binding — or doesn't. **Take-or-pay** obligates you to pay for capacity whether you draw it or not; its cousins are the wafer bank and the NCNR (non-cancellable, non-reschedulable) window. Whether these protect you or trap you depends entirely on which seat you're in.

Who carries it. Usually the buyer — but rarely symmetrically, and the asymmetry runs both ways depending on leverage. SkyWater's anchor customer commits to *"order commitments not less than: (i) 95% for the first Calendar Month … (ii) 90% for the second"* of its own forecast. Everspin gave GlobalFoundries a *"binding volume commitment."* Allegro is bound to a *"six month binding forecast"* to its captive foundry. Mobileye's orders to Intel are *"non-cancellable and non-reschedulable ('NCNR'), at a minimum, during the 6 months lead time"* — and *"Intel may … change the NCNR period in its sole discretion."* Yet Aquantia's buyer, Intel, committed to **nothing**: *"forecasts … do not constitute a … commitment by Buyer."* Same clause family, opposite risk allocation.

Advice. Find the *binding window* inside the "non-binding" forecast — that is your real exposure, and it is often not where the headline number sits. If you're the buyer, model take-or-pay against a **demand-down** scenario, not the plan; that's when it bites. If a counterparty can move the NCNR window "in its sole discretion," your commitment is open-ended in disguise. The term you want is a **symmetric** one, and you rarely get it without asking.

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2 · Supply continuity & allocation — the clause you only read in a crisis

The idiosyncrasy. Continuity is rarely lost in the supply clause. It's lost in **force majeure** and **allocation** — the fine print that decides, when there isn't enough to go around, whether *your* line keeps running. The peculiarly semiconductor move: the supplier may suspend delivery for events beyond its control while the buyer's payment and commitment obligations continue. Add single-sourcing — common, because qualification is expensive and slow — and one plant, one ministry, one export order can stop your supply.

Who carries it. The buyer, structurally — force-majeure suspension rights run to the party who makes the goods, and allocation-in-shortage is written by the party who's short. GlobalFoundries is unusually candid that this cuts upstream too: *"GLOBALFOUNDRIES' operations depend upon the continuous and uninterrupted supply of Products … GLOBALFOUNDRIES could suffer significant damage"* — from a single French substrate maker. Mobileye single-sources its EyeQ processor to STMicro and its photonics to Intel. And the Nexperia case is the pure form: a single-sourced part, no contractual second-source right, and a supply that stopped when one government acted.

Advice. Read the force-majeure clause as a **continuity clause**, because that's what it is. Ask three questions: on a shortage, what is my **allocation priority**? Do I have a contractual **second-source / right-to-qualify**? Does my payment obligation survive a suspension I didn't cause? If the first two are silent, you are single-threaded whether or not anyone said so.

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3 · Quality & yield — the protection buyers want most and write down least

The idiosyncrasy. On a mature node, yield is the supplier's problem. On a new node it's a negotiation — and the contract decides who eats the scrap. The semiconductor-specific instruments are the **yield floor** (a guaranteed good-die rate), the remedy ladder (*rescreen → repair → replace → refund*, or respins/credits), and the **epidemic / systemic-defect** clause for a failure that runs across lots. These move real money and are the terms most often left vague or absent.

Who carries it. Whoever failed to specify it — usually the buyer, because the default is that risk sits with the party holding the parts. Aquantia's remedy tops out at *rescreen / repair / replace / refund* with **no yield floor defined at all**. Where a supplier does take it on, it's narrow and explicit — Mobileye's foundry agreement makes *"ST … responsible for failures due to manufacturing, testing and assembly,"* and no further.

Advice. If you're buying a part on an immature process, a missing yield floor is not a neutral silence — it's a risk you've accepted by default. Ask for a defined floor **and** a defined remedy, and pin down who bears the cost of a systemic defect found after acceptance. If you're the supplier, an undefined floor cuts against *you* in a dispute — precision protects both seats.

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4 · Liability & indemnity — the cap that only protects one side

The idiosyncrasy. Everyone caps liability. The semiconductor tell is **asymmetry**: the supplier caps its exposure at a small, knowable number while the buyer's *commitment* runs open-ended — so the two sides of the same contract live under completely different ceilings. The consequential-damages waiver is near-universal and looks mutual; the *aggregate* cap almost never is.

Who carries it. The buyer of a capped supplier's parts, and the weaker party generally. SiTime's foundry hard-caps itself at *"$5,000,000 (five million U.S. dollars)"* — and SiTime separately agrees to *"defend … and hold Bosch … harmless."* Everspin's foundry caps at *"THE TOTAL PURCHASE PRICE RECEIVED BY GLOBALFOUNDRIES FROM CUSTOMER IN THE TWELVE (12) MONTHS"* preceding a claim, while Everspin carries the binding take-or-pay. Allegro's foundry caps *"only on the supplier's side."* And Cerebras couldn't even see its ceiling — the filed term is *"Liability Cap [***]."*

Advice. Never read the cap in isolation — read it against your **own** open-ended obligations in the same contract. A $5M supplier cap against a multi-year take-or-pay is not risk-sharing; it's a one-way door. Ask whether the cap is mutual, whether it excludes the things that actually hurt (IP indemnity, epidemic defect, confidentiality breach), and whether "fees paid in 12 months" is anywhere near your real downside. A redacted cap is itself a finding: you are being asked to accept a limit you're not allowed to see.

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5 · IP ownership & residuals — "owned" and "licensed" are different assets

The idiosyncrasy. In chips the IP is frequently **rented, not owned**, and the contract governs a two-way flow of know-how the parties can't fully separate. Hence the peculiar instruments: the **process-learning firewall**, the **improvement assignment** (you hand your own improvements *back*), and the **residuals** clause (unaided memory walks out the door). The other party's trade secrets can be embedded in the parts you buy.

Who carries it. The party that mistook a license for ownership. MetaOptics licenses its **entire** technology base — *"a non-exclusive, non sublicensable, non-transferable, revocable for cause licence"* — from a Singapore state agency.

SkyWater's manufacturing process is licensed from its former parent and its own improvements flow back: *"Licensee hereby irrevocably transfers, conveys, and assigns to Licensor in perpetuity all … Licensor Improvements."* In Mobileye's most strategic collaboration, *"Intel solely owns all Project IPR to Intel Core Technology."* Aquantia ships parts in which *"the Mask Works may include trade secrets of Intel."* Allegro's IP is *"jointly owned by Sanken and Allegro."*

Advice. Separate what you **own** from what you **license**, and treat *non-exclusive / revocable* as a leash, not an asset — an acquirer or investor will. Read improvement-assignment and residuals clauses as ownership questions, not boilerplate: they decide whether the value you create accrues to you or to your counterparty.

6 · Change of control & ownership — the exposure nobody signs

The idiosyncrasy — and the one generic review misses entirely. The contract is typically **silent** on who *ultimately owns* your counterparty: no disclosure obligation, no change-of-control right, nothing that fires when the parent behind your supplier changes. So control can flip — a foreign acquisition, a parent crossing below majority — without a single word changing in your agreement. This is the Nexperia dimension, and its lesson is the **two clocks: the event takes a month; the exposure takes years.** The seizure, the export block, the parts not arriving — that's the fast clock, and nobody forecasts it. Being single-sourced on a supplier whose ultimate owner you cannot see, under a contract with no disclosure duty and no change-of-control right — that's the slow clock, and it's been visible the whole time. *Nobody forecasts the lightning. Anybody can notice they've been standing in dry brush.*

Who carries it. The buyer, always — the ownership of your supply base is your exposure, and the contract wasn't written to defend it. When the tripwire *is* written in, it detonates on schedule: Allegro's rights *"shall terminate in the event that Sanken ceases to own, directly or indirectly, more than fifty percent"* — Sanken crossed that line and Allegro was put in play. Mobileye's contractual rights hinge on Intel holding *"80.1% ownership."* Cerebras's IPO was frozen by a CFIUS review of G42's stake. GlobalFoundries, a *"controlled company"* of Abu Dhabi's Mubadala, *"removed … corporate opportunity doctrine provisions"* — pre-consenting to its owner's conflicts.

Advice. Add two clauses you probably don't have: a **beneficial-ownership disclosure** obligation and a **change-of-control right**. And screen the **owner**, not just the counterparty — the name on the contract can be clean while the parent is not. At scale, don't try to know every owner; know **which of your suppliers you cannot resolve an owner for**, and spend your diligence there. That short list is the whole game.

7 · Exclusivity & exit — the second source you gave away by contract

The idiosyncrasy. Your second source can be foreclosed by **contract**, not by cost or capability. Exclusivity clauses bar you from qualifying an alternative; facility-restriction clauses lock a process to one fab. And exit is asymmetric: one side can terminate for convenience, the other cannot — so "flexibility" is a right only one party holds.

Who carries it. The party who signed away optionality. Everspin *"shall not disclose, dispose of or license any Foreground IP to any foundry"* but GlobalFoundries — no second foundry, by contract. SkyWater's process *"may be used … only at the Facility."* Aquantia's counterparty *"may terminate … for convenience at any time,"* while *"Aquantia may not."* SiTime can leave Bosch only *"upon providing three (3) years advance written notice."* MetaOptics' entire license base is *"revocable for cause"* on a missed commercialization milestone with *"3 months' notice."*

Advice. Price exclusivity as the **option you're surrendering** — a second source you can't qualify is a cost even if you never use it. Diary the **initial-term expiry and renewal windows**: they are your scheduled moments of leverage, and they pass unused if no one is watching. If termination-for-convenience runs one way, know which way, and what it does to your committed volume.

8 · Price — you can't see the level, but you can see the mechanism

The idiosyncrasy. Price is the term most reliably redacted — and the least important structurally. What survives redaction is the **mechanism**: a ceiling schedule, an escalation formula, a most-favored-nation clause, a fixed-for-N-months lock. Aquantia's agreement, for instance, holds price *"under … the price ceiling or price reduction schedule set forth on Addendum A for a period of 18 months."* The rule is visible even when the number is `[***]`.

Advice. Don't chase the redacted number — read the **rule** that moves it. A fixed schedule, a formula tied to an index, and an open renegotiation are three very different risk profiles at the same headline price. And notice when a cap *is* a hard number rather than a redaction (SiTime's $5M) — a stated figure is a negotiated position you can benchmark; a redaction is one you can't.

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9 · Warranty & infringement — the protection that gets disclaimed

The idiosyncrasy. The clauses that look like fine print — warranty disclaimers and IP-infringement indemnity — quietly decide who owns the tail risk. In semiconductors the licensor or supplier routinely disclaims the two things a buyer most needs: that the technology *works* for the purpose, and that it *doesn't infringe someone else's IP*.

Who carries it. The licensee/buyer, by disclaimer. MetaOptics' state licensor *"MAKES NO WARRANTY OR REPRESENTATION AS TO THE VALIDITY OR SCOPE OF THE TECHNOLOGY, OR THAT THE TECHNOLOGY WILL BE FREE FROM INFRINGEMENT"* — so a third-party claim on the licensed metalens IP is MetaOptics' problem, while the licensor's own liability caps at *"any amount that LICENSEE has actually paid."* Mobileye's Intel cross-license disclaims *"all implied warranties of merchantability, fitness for a particular purpose, and non-infringement."*

Advice. Find the infringement-indemnity clause and read who defends whom. If the party licensing you the technology won't warrant it's non-infringing, *you* are underwriting its patent risk — price that, or push it back. A disclaimer of "validity or scope" on your core technology is a diligence red flag, not boilerplate.

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10 · The marquee-customer illusion — a logo is not a commitment

The idiosyncrasy. A famous customer on the pitch deck can hide *total* dependence, and — separately — can owe you nothing. Semiconductor revenue is peculiarly lumpy: a handful of design wins can be most of the book, often with **no long-term agreement** behind them, so "revenue" and "committed revenue" are very different numbers.

Who carries it. The supplier who mistook concentration for security. Mobileye's three largest customers — *"35%, 19%, and 17% of revenue"* — had **no long-term written agreements** (a fact the SEC forced into the prospectus). Cerebras filed **eleven** supply agreements and every one is with G42 or its orbit — a single customer that is also its investor and its lender. Aquantia's biggest relationship, Intel, could *"terminate … for convenience"* and committed to no volume — the marquee customer was the one it least controlled.

Advice. Separate revenue from **committed** revenue: how much rides on customers who could leave tomorrow with no penalty? Concentration in a customer who is also your investor/lender is a governance exposure, not just a sales one. "We supply [famous company]" often means "our biggest relationship is the one we have the least control over" — test which it is.

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11 · Governing law, forum & enforcement — where you'd actually have to sue

The idiosyncrasy. With a foreign counterparty, the enforcement terms decide whether your rights are real. Governing law, arbitral forum, and — increasingly — **territory-specific enforcement carve-outs** determine where and whether you can actually collect. Chip companies now actively engineer where their IP can be enforced.

Who carries it. The party who'd have to enforce abroad. SiTime's 2023 acquisition structured its licenses around *"Exclusive Licenses"* with **China-territory enforcement carve-outs** — a company deliberately managing where its IP is defensible. SkyWater's national-security amendment bars assignment *"to any legal entity not organized under the laws of the United*

States" without consent — jurisdiction written directly into the deal.

Advice. Check governing law and forum against where the counterparty's **assets** actually are — a Delaware judgment against a counterparty whose value sits offshore may be unenforceable. Treat territory-specific IP carve-outs as a map of where your rights stop. For a cross-border supply relationship, "where would I sue, and could I collect?" is a term, not an afterthought.

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12 · Export control & geopolitics — table stakes, with a twist

The idiosyncrasy. Export-control language (EAR/ITAR, responsibility assignment) is **universal** — it's in essentially every agreement, so its *presence* differentiates nothing and every buyer already monitors it. The twist: the boilerplate captures the wrong risk. The real geopolitical exposure isn't whether the clause exists — it's **who owns the counterparty and where the technology flows.** Nexperia's contracts had clean export language; the exposure ran through **Wingtech**, the parent, which the export clause never mentioned. SkyWater's "Trusted Foundry" status is *clauses* — CUI/CDI/FOUO handling, US-entity assignment limits. MetaOptics' licenses restrict the technology to *"commercial and/or civilian purposes only."* Cerebras's hardware moves US→UAE, with delivery *"contingent on government approval."*

Advice. Treat the export clause as hygiene — necessary and insufficient. The exposure that surprises you lives one level up, in **ownership** (dimension 6), and one level out, in where the licensed technology can travel. Don't over-index on the clause everyone already reads; spend your attention on the ownership nexus behind it.

The through-line

Across every dimension the same thing is true: **risk allocation is rarely symmetric, and the weaker party signs it without pricing it.** The exposures that matter most — the binding window, the allocation priority, the missing yield floor, the one-sided cap, the disclaimed warranty, the uncommitted marquee customer, the unseen parent — are not hidden. They are **written down, and unread.** That is not a limitation of the paper. It is the entire opportunity.

We can run this read on your own agreement. Send one representative foundry, customer, or license contract under mutual NDA, and we'll walk it dimension by dimension against the idiosyncrasies above — who carries which risk, and what to ask for before you sign.

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